

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe
Software Engineering Intern

San Jose, CA
June 2026 – Sept. 2026

- Designed a time-series anomaly detector in Python, selecting a machine learning model to handle multi-layer seasonality across **10+** workflows and **30+** countries.
- Automated an MLOps pipeline for weekly retraining and hourly inference of forecasting models.
- Profiled **60.6M+** payment events in Databricks, isolating **86,387** duplicates and a **47%** null rate.

VALT Health
Founding Software Engineer

Remote
Aug. 2025 – Feb. 2026

- Engineered a Python Flask backend to ingest **100k+** datapoints from fitness devices, designing schemas to unify data formats across **20+** distinct sources.
- Implemented async batch processing in GCP, raising execution speed **35%** and cutting writes **60%**.
- Automated local environment setup and data seeding via Bash, cutting testing time by **80%**.

RESEARCH

Graph Theory & Neural Architecture
Undergraduate Research Fellow (NSF REU)

San Diego, CA
March 2026 – Present

- Engineer a Python pipeline to analyze LLM embedding manifolds (Llama-2, DeepSeek-V2), building utilities that compress **500GB** checkpoints into **13GB** localized manifolds.
- Optimize placement scripts and preprocessing for a **30x runtime reduction** on **8GB RAM** systems.
- Apply **Rent's Rule** and Louvain clustering to identify model backbones, discovering that noise-filtering raises Inter-Cluster Network density from **0.18%** to **13.25%** in Llama-2-70B.

Applications of Multi-Agent Systems
Undergraduate Research Fellow (UC Scholars)

San Diego, CA
June 2025 – Aug. 2025

Multi-agent AI Chess Tutor | *Python, MCP, Django, React, SQLite*

- Developed **7 REST API endpoints** in Django powering a multi-agent AI chess tutor for **15+** players.
- Architected multi-agent workflows cutting hallucinations **80%** and accelerating runtime **83%**.
- Engineered custom **MCP** servers and tools standardizing tool access across all agents.

PROJECTS

NanoGPT | *Python, PyTorch, Transformers, Neural Networks* | [🔗](#)

Mar. 2026

- Implemented a transformer from scratch in **Python** and **PyTorch**, building multi-headed attention and backpropagation to train a **100k+** parameter character-level model.
- Engineered scalable dataloaders streaming tokenized datasets at **100%** GPU utilization.

SquaredUpChess | *React, TypeScript, Django, Websockets, AWS, Coolify, Docker* | [🔗](#)

Aug. 2025 – Dec. 2025

- Architected **20+ Django REST API endpoints** for authentication, sessions, and transactions.
- Established a **CI/CD pipeline** with automated testing and **Docker** deploys at **99.9%** reliability.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, SQL (Postgres/SQLite), Java, JavaScript, Bash, HTML/CSS

Development: RESTful APIs, Django, FastAPI, Flask, Node.js, React, Asyncio, WebSockets, Data Structures & Algorithms, CI/CD, PyTest

AI & Cloud: PyTorch, TensorFlow, Transformers, LLMs, NLP, Machine Learning, Multi-Agent Systems, MCP, AWS, GCP, Docker, GitHub Actions, Linux